DENVar: A Global Dynamic Estimate of NDVI Variance

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Abstract

Remote-sensing proxies of habitat productivity such as the Normalized Difference Vegetation Index (NDVI) have helped develop and support many hypotheses and projects. However, historically, users tended to focus solely on trends in mean productivity. In recent years, the variance around the mean (i.e., stochasticity) has gained more attention, particularly as climate change and human-induced rapid environmental change transform the ecosystems species have evolved in and adapted to. Still, calculating trends in NDVI stochasticity requires careful modeling, since stochasticity estimates are conditional on the estimated mean and the scale at which the data are collected and modeled. We thus present a new global Dynamic Estimate of NDVI Variance, *DENVar*. We estimate spatiotemporal trends in NDVI mean and variance using Generalized Additive Models, which provide a flexible yet transparent and interpretable model structure. We show that DENVar varies nonlinearly with ***XXX*** and can be used to test many hepotheses related to environmental stochasitcity, such as optimal foraging theory and regime shifts as well as for conservation. We provide a publicly and freely available repository for downloading rasters of daily and long-term averages of DENVar and the related estimates of mean NDVI.

1 Introduction

The Normalized Difference Vegetation Index (NDVI) is a proxy of primary productivity used in a wide range of disciplines, including ecology (Pettorelli et al., 2005, 2011), landscape phenology (Merkle et al., 2016; Hmimina et al., 2013), agriculture (Franch et al., 2017), forestry (João et al., 2018), and conservation (Mallegowda et al., 2015; Marcus et al., 2025). High-resolution, remotely-sensed NDVI data have allowed analysts to estimate plant biomass without fieldwork (Bhatti et al., 2024; Borowik et al., 2013) and leverage such estimates for studying large-scale plant dynamics (Pettorelli et al., 2005) and their effects on animal movement behavior (Merkle et al., 2016; Aikens et al., 2017). Still, most NDVI work is limited to quantifying changes in and responses to mean NDVI while ignoring changes in the variance around the mean.

Environmental stochasticity has been studied for decades (Levins, 1974; Stephens and Charnov, 1982; Lande, 1993; Sæther, 1997), but recent work has drawn more attention to the effects environmental stochasticity has at a wide range of scales, from individual organisms (Mezzini et al., 2025; Rizzuto et al., 2021) to landscapes (Zhang et al., 2009; Mutti et al., 2020; Marcus et al., 2025). However, estimating trends in NDVI variance requires more nuanced modeling than estimating changes in mean NDVI, particularly when environments are temporally dynamic and spatially heterogeneous. The scale at which processes are observed determine how individuals' perception of and ability to respond to change (Levin, 1992). A forest fire is severely disruptive and deadly for many insects (Fettig et al., 2022), while birds and large mammal may not struggle to escape the fire and or establish a new range (Franklin et al., 2021; Calhoun et al., 2024). Similarly, the effects of the fire recovery will depend on its duration relative to other organisms' phenology and lifespan (Frankham and Brook, 2004). Therefore, distinguishing between a change in mean NDVI and a stochastic fluctuation requires a conscious decision about the scale at which to consider the data.

Previous work suggests that environmental stochasticity can reduce limit species' ability to adapt or specialize (Chevin et al., 2010), alter species' movement behavior (Mezzini et al., 2025), and even affect extinction risk at both the population and species levels (Lande, 1993; Bull et al., 2007; Fung et al., 2018). Still, the effects of environmental stochasticity on species dynamics and ecosystem stability remain understudied (Marcus and Noonan, 2023). Global, temporally dynamic estimates of NDVI variance would facilitate research by building upon the widespread use and interpretability of NDVI wile also removing the barriers of computational demand and statistical knowledge that are required for quantifying environmental stochasticity. However, since NDVI is not a direct measurement primary productivity but a proxy, it should be interpreted carefully (Pettorelli et al., 2011). NDVI is prone to sensitivity loss in very green environments (Matsushita et al., 2007; Son et al., 2014), values vary across sensors (Beck et al., 2011; Fan and Liu, 2016; Huang et al., 2021), and the coarse spatiotemporal scale of satellite-derived NDVI can be a substantial barrier when assessing fine-scale trends (Goodchild and Quattrochi, 2023; Reid et al., 2018). However, careful statistical modeling can help overcome some of these limitations. In particular, Generalized Additive Models (GAMs) are particulalrly convenient due to their ability to learn flexibly from data (Wood, 2017; Simpson, 2018) and handle very large spatiotemporal datasets (Wood et al., 2015, 2017).

We present a new measure of environmental stochasticity, which we name the Dynamic Estimate of NDVI Variance (DENVar). We calculate DENVar using 44 years of NDVI data from the AVHRR and VIIRS collection of sensors and GAMs. Although satellites with AVHRR and VIIRS sensors provide data at a coarser spatial resolution (0.05°) than other satellites (e.g., 250m for MODIS), they are the only sensors to provide data at a daily frequency and and for more than 40 years. Our use of GAMs provides a flexible yet interpretable approach for estimating trends in mean and variance in NDVI without the rigidity of other more traditional methods (e.g., ARIMA) or the opaqueness of machine learning models such as neural networks (Simpson, 2018; Wood, 2017). We illustrate the relationships between DENVar and multiple, relevant ecological metrics, and we provide access to a free repository of daily, global estimates of DENVar and the corresponding estimates of mean NDVI.

2 Methods

2.1 Choice of color palettes

We represent NDVI using a modified version of Fabio Crameri's divergent *bukavu* palette (Crameri, 2018a; Crameri et al., 2020), which has high-contrast for deuteranope and protanope vision. For NDVI values between -0.1 and 1, the colors also have sufficient contrast for the colors to also be distinguishable by strong tritanope and achomatic vision. Fig. A.1 contains an approximate representation of how each vision type would perceive each color palettes used. We obtained all color palettes with three or more colors from the `khroma` package (v. 1.14.0 Frerebeau, 2024) for `R` (v. 4.4.0; R Core Team, 2025).

2.2 Datasets used

Fig. 1 illustrates the processes of data collection, cleaning, and modeling used in this project. We downloaded NDVI image composites directly from the servers for data from AVHRR sensors (1981-06-24 to 2013-12-31; United States National Centers for Environmental Information (NCEI) and United States National Aeronautics and Space Administration, 2025; Vermote and NOAA CDR Program, 2018) and VIIRS sensors (2014-01-01 to 2025-06-29; United States National Aeronautics and Space Administration, 2025; Vermote and NOAA CDR Program, 2022).

We removed all pixels labeled as "cloudy" for AVHRR rasters and "probably cloudy" or "confidently cloudy" for VIIRS ratsers. To remove pixels that included large portions of water, we used a shapefile of major water bodies (Garmin International, Inc. and Esri, 2025) to produce a static raster of the percentage of water within each pixel, at the same spatial resolution as the NDVI rasters (0.05°; Fig. A.2). We removed all pixels with $>$ 40% water cover.

Many rasters had horizontal bands of NDVI $>$ 0.9 at unrealistically northern latitudes between September and March, likely due to limited visibility during fall and winter in the Northen hemisphere. We identified and removed problematic pixels using the monthly empirical cumulative distribution functions of NDVI across latitude (Fig. A.3). We did not filter problematic values at latitudes $<$ 60°S since we fully excluded Antarctica from the analyses (Fig. A.4).

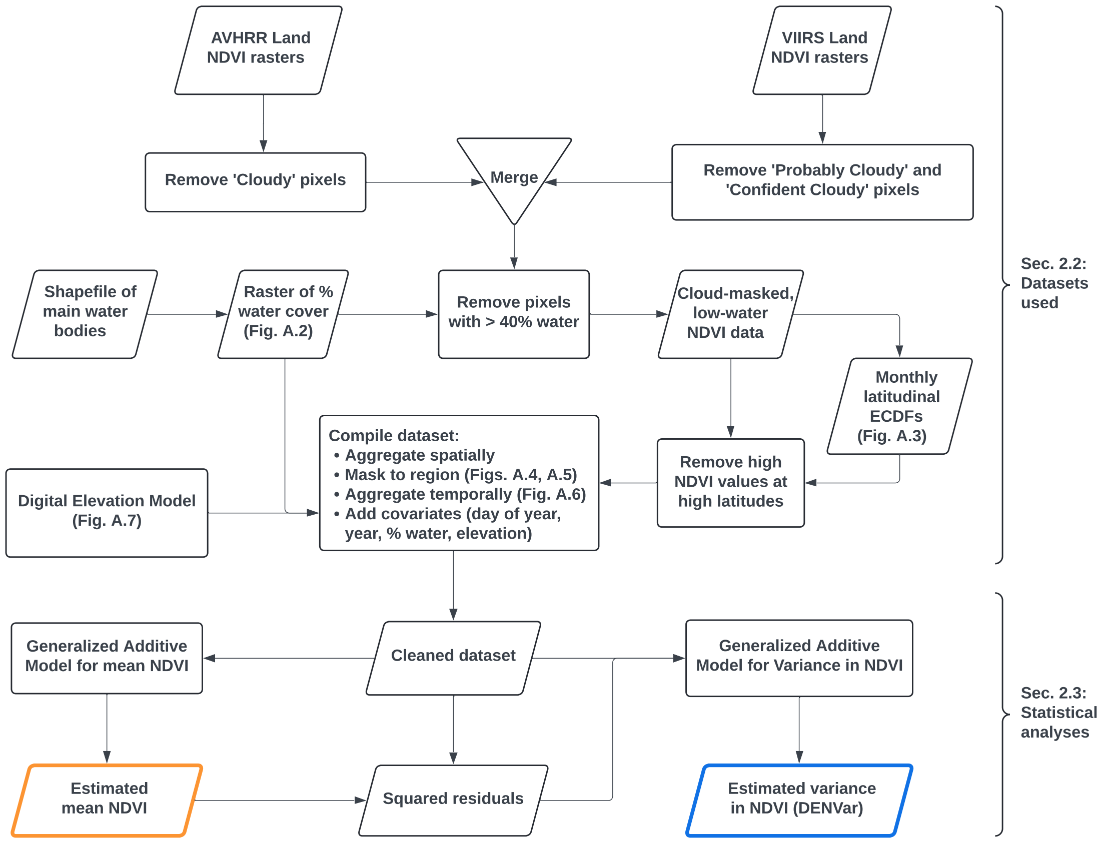

Figure 1: **Flowchart for the analyses used in the manuscript.** The curly braces on the right indicate the subsections in the Methods section that detail the dataset collation and cleaning (2.2), and statistical analyses (2.3).

To ensure the number of rows for each dataset was below the maximum data frame size in R ($3.17 \times 10^{10} > 2^{31} - 1 \approx 0.21 \times 10^{10}$), we reduced the dataset size by splitting it across four spatial groups (Figs. A.4 and A.5) and spatially and temporally aggregated the rasters by a factor of four. The final dataset thus had one raster every four days with a spatial resolution of $0.05° \times 4 = 0.2°$ (Fig. A.6). Finally, we annotated the dataset with columns of day of year (1-366), integer year (1981-2025), percent water cover, and elevation above sea level (m). The digital eleveation model was obtained using the `get_elev_raster()` function from the `elevatr` package for R (v. 0.99.0; Hollister et al., 2023) with an original resolution of 0.076°(Fig. A.6). Despite the substantial reduction in dataset size, the dataset creation was still challenging for the machine used, which had 2.2 TB of RAM and two Intel Xeon Platinum 8462Y+ processors (64 cores total).

2.3 Statistical Analyses

We estimated spatiotemporal trends in mean and variance in NDVI using Generalized Additive Models (GAMs) via the `mgcv` package for R (v. 1.9-3 Wood, 2017). To reduce modeling fitting times with negligible losses to model accuracy, we used the `bam()` function with fast REstricted Marginal Likelihood (`method = fREML`) and covariate discretization (`discrete = TRUE`; Wood et al., 2015, 2017). Ideally, NDVI should be modeled using beta location-scale models (after the linear transformation $Y^* = (Y+1)/2$) to account for the fact that: (1) NDVI is bounded between -1 and 1, and (2) the variance in NDVI is dependent on the mean (and vice-versa), since ecosystems with very low NDVI (e.g., rock, ice, or concrete) or very high NDVI (e.g., dense forest) tend to have lower variance in NDVI. However, preliminary tests and work by (Marcus et al., 2025) showed that beta GAMs and particularly beta location-scale GAMs had prohibitive requirements for both fitting times and memory requirements, since both requirements scale $O(Mnp^2)$, for M smoothing parameters and a model matrix with n rows and p columns (Wood et al., 2015). A beta location-scale model for only the island of Sardigna (24,000 km^2; Appendix B) was interrupted after 144 hours, during which the model had only completed three steps using the Newton method described by Wood and Fasiolo (2017). In contrast, Gaussian models with fixed scaled parameters (i.e., constant variance) fit substantially faster and provided very similar spatialtemporal estimates of mean and variance in NDVI (Appendix B). Thus, we estimated trends in mean and variance in NDVI using a pair of GAMs for each region shown in Fig. A.4; one for estimating mean NDVI over space and time (assuming a constant variance), and one for estimating the mean squared residual (i.e., variance) in NDVI (again, assuming a constant scale parameter).

Together, the sets of models for mean NDVI (i.e., μ) and and variance in NDVI (i.e., σ^2) had the following structure, with ν indicating NDVI:

$$\begin{cases} \nu \sim \text{Normal}(\mu, \sigma^2) \\[2em] \mu = \mathbb{E}(\nu) \approx \hat{\mathbb{E}}(\nu) = \hat{\mu} \\ \hat{\mu} = \beta_\mu + s(\texttt{prop\_water}) + s(\texttt{elev\_m}) + s(\texttt{doy}) + s(\texttt{year}) + s(\texttt{lat}, \texttt{long}) + \\ \qquad t(\texttt{doy}, \texttt{year}) + t(\texttt{doy}, \texttt{lat}, \texttt{long}) + t(\texttt{year}, \texttt{lat}, \texttt{long}) \\[2em] \sigma^2 = \mathbb{E}\left((\nu - \mu)^2\right) \approx \hat{\mathbb{E}}\left((\nu - \hat{\mu})^2\right) = \hat{\sigma}^2 \\ \hat{\sigma}^2 = \beta_{\sigma^2} + s(\texttt{prop\_water}) + s(\texttt{elev\_m}) + s(\texttt{doy}) + s(\texttt{year}) + s(\texttt{lat}, \texttt{long}) + \\ \qquad t(\texttt{doy}, \texttt{year}) + t(\texttt{doy}, \texttt{lat}, \texttt{long}) + t(\texttt{year}, \texttt{lat}, \texttt{long}) + s(\hat{\mu}) \end{cases}, \tag{1}$$

where β_μ and β_{σ^2} are model intercepts, and, for a given (spatiotemporally aggregated) pixel: `prop_water` is the proportion of water within the pixel, `elev_m` is elevation above sea level in meters, `doy` is integer day of year (1-366), `year` is integer year (1981-2025), and `lat` and `long` are latitude and longitude, respectively. We omit the pixel-level indices for ease of readability. $s(\ )$ indicates marginal smooth functions, while $t(\ )$ indicates tensor product interaction terms. Seasonal terms used cyclical cubic splines (`bs = 'cc'`) with edge knots 0.5 and 366.5, while spatial smooths used a basis with second-order spline on the sphere (`bs = 'sos'`; Wood, 2017). The linear predictors for μ and σ^2 have the same structure, other than the $s(\mu)$ smooth term for σ^2, but the functions for σ^2 are independent of those for μ. Despite the data aggregation procedures and faster fitting methods, including parallelizing computation across 60 cores, each of the four models required multiple days to fit. Predicting from the models for $\hat{\mu}$ was also had substnatial time and memory demands, since predictions were calculated without discrete prediction methods (`discrete = FALSE`). While discrete predictions were orders of magnitude faster, they were excessively spatially coarse. Predicting fitted values required 40 dats for the Neotropic-Nearctic region despite consistently using $>$ 1 TB of RAM. Predictions for the other three regions were ran with covariate discretization. See Appendix C and the code availability statement for additional information.

From the two sets of models, we produced daily estimates of (1) the Dynamic Estimate of NDVI Variance (DENVar) and (2) the estimated mean NDVI they are conditional on, at a ***XXX*** °spatial scale. The rasters are freely available on GitHub (see code availability statement), along with long-term (1981-2025) averages of both DENVar and the estimated mean NDVI. Since there are no corrections to any bias of $\hat{\mathbb{E}}\left((\nu - \hat{\mu})^2\right)$, the estimates of σ^2 may be inflated near areas or periods of rapid and substantial change. Still, the appreciability of this inflation depends on the spatiotemporal scale over which changes are perceived (Levin, 1992; Frankham and Brook, 2004; Goodchild and Quattrochi, 2023; Mezzini et al., 2025). Since estimates of variance are conditional on the corresponding estimates of the mean, we suggest users always use the corresponding rasters of mean NDVI in their analyses rather than using independent estimates of mean NDVI. DATA CAN BE DOWNLOADED VIA XXX.

3 Results

The estimates of mean NDVI ($\hat{\mu}$) and DENVar ($\hat{\sigma}^2$) were spatially smooth but still captured complex spatial trends (Fig. 2). The models for mean NDVI explained 68-88% of the deviance, while the models for DENVar explained 13.40-18.60% of the deviance (Table refdev-expl-table). For both models, predictions were unreliable in areas with very scarce data (e.g., rock and ice biomes in the Arctic; see Fig. A.5), so such areas were excluded when producing figures and estimates of mean NDVI and DENVar. Still, the models detected complex spatial patterns in both variables, such as along the Andes and the Himalayas, and in the Palearctic and Nearctic taiga.

Region	Dev. expl. (Mean NDVI, %)	Dev. expl. (DENVar, %)
Africa	88.02	18.60
Indo-Malay, Oceania, Australasia	67.89	18.17
Neotropic, Nearctic	80.65	13.40
Palearctic	82.23	14.89

Table 1: Percent deviance explained by the models for mean NDVI and DENVar for each of the four regions shown in Fig. A.4.

The models also detected complex spatiotemporal trends, with major shifts occurring proximally to the major El Niño events in 1982, 1997, and 2015 (Fig. 3; Jia et al., 2025). The early 1980s saw lower mean NDVI and greater variance in NDVI, with values increasing during the second half of the decade. In the 1990s, there were two global-scale, multiannual oscillations in both mean NDVI and DENVar that ended with a widespread increase in mean NDVI and NDVI stochasticity in 2000 and 2001. The 2000s and 2010s were characterized by more heterogeneous and complex spatial patterns that intensified just before the 2015 El Niño event. Equatorial regions saw a decrease in mean NDVI and DENvar, while near-tropical regions generally experienced an increase in both metrics. Latitudes approximately above 60° experienced the strongest changes in both mean NDVI and DENVar, with strong spatiotemporal trend inversions near 59, 65, and 69°N.

Spatial patterns in long-term mean NDVI and DENVar varied across biomes (Fig. 4), with many biomes showing a wide range of both variables. However, DENVar and mean NDVI often had a concave down relationship with a peak in DENVar near intermediate values of NDVI (0.2-0.6).

Fig. 5 shows the relationship between DENVar and some common spatial variables. Generally, DENVar: was highest with mean NDVI near 0.25, decreased with mean annual temperature, increased with seasonal temperature range, and increased with total annual precipitation. Additionally, DEVNVar was generally higher in environments with low: IUCN Red List species richness, proporition burned, human footprint, and risk of arbivoral disease.

4 Discussion

We have presented a new spatiotemporally dynamic metric of environmental stochasticity, which we have called the Dynamic Estimate of NDVI Variance, DENVar. We have shown that DENVar has a strong, nonlinear relationship with with multiple crucial indices, including species diversity, proportion of area burned, and anthropogenic alteration of the enviroment. Yet, environmental stochasticity is rarely accounted for in conservation research (Marcus and Noonan, 2023), although recent work has started drawing more attention to how environmental stochasticity shapes ecosystems and evolution (Anderson et al., 2017; Logares and Nuñez, 2012).

Importantly, environmental stochasticity should not be confused with seasonality. Seasonality is generally best interpreted as a (cyclical) change in mean conditions that many organisms can learn to predict, as in the case of green-wave surfing (Middleton et al., 2018; Geremia et al., 2019). In contrast, environmental stochasticity refers to unpredictable changes, such as fires (Fettig et al., 2022; Calhoun et al., 2024), sudden snow storms (Aikens et al., 2025), and unusual seasonal phenology (Torstenson and Shaw, 2025). Riotte-Lambert and Matthiopoulos (2020) and Mezzini et al. (2025) provide a theoretical framework for how and why resource stochasticity increases organisms' range sizes (also see Stephens and Charnov, 1982; Rizzuto et al., 2021). Highly unpredictable enviroments tend to have lower energetic balances and carrying capacities (Chevin et al., 2010) and higher extinction risk (Lande, 1993; Fung et al., 2018), since communities cannot rely on stable conditions. Instead, stochastic environments reduce species' ability to respond to change and specialize, favoring generalist and flexible adaptations instead (Levins, 1974). In memory is often crucial for navigating highly stochastic environments (Steixner-Kumar and Gläscher, 2020), including locating dependable waterholes during droughts (Polansky et al., 2015), avoiding ecological traps (Abrahms et al., 2019), and the development of home ranges (Falcón-Cortés et al., 2021; Ranc et al., 2022) and migration (Bracis and Mueller, 2017).

Failing to account for environmental stochasticity when designating protected areas can result limit the strategies' effectiveness. Stochastic environments cause animals to require larger ranges

(Mezzini et al., 2025) and alter the timing and path of their migration (Ortega et al., 2023; Aikens et al., 2020). Therefore, environmental stochasticity should be accounted for when evaluating current protected areas and designating new ones, but lack of research on environmental stochasticity in conservation suggests that protected areas may have have similar levels of environmental stochasticity to unprotected areas. Marcus et al. (2025) supports this speculation and also shows that that more stochastic environments tended to have lower species diversity and a greater frequency of months extreme mean temperatures. Their findings are particularly concerning given the ubiquitously rapid rate of climate change (Loarie et al., 2009; Diffenbaugh and Field, 2013; IPCC, 2023), icluding the increased frequency and severity of extreme events (IPCC, 2023), and the increasing anthropogenic pressure on wildlife (Tucker et al., 2018; Weststrate et al., 2024). Each of these threats is then amplified by the difficulty in responding to multifarious change (i.e., concurrent change across in multiple variables; Polazzo et al., 2024), especially for species with strong site fidelity (Merkle et al., 2022). For example, Aikens et al. (2022) found that industrial activity reduced the ability of mule deer (*Odocoileus hemionus*) to migrate following forage availability, while (Sawyer et al., 2019) showed that mule deer's strong site fidelity prevents them from adapting to changing environments. In contrast, Ortega et al. (2023) demonstrated that mule deer adapt their migration to changes in phenology, suggesting that adaptability is likely population-dependent.

Quantifying trends in variance can also be helpful in broader disciplines, such as macroecology and landscape ecology. Changes in habitat variance over time can be indicators of ecological transitions and regime shifts (Carpenter and Brock, 2006; Bjorndahl et al., 2022), since increases in stochasticity may push systems into an alternative stable state, whether such systems are habitats (Hastings et al., 2021) or species undergoing evolutionary tipping points (Botero et al., 2015). Furthermore, while our work is limted to terrestrial systems, similar approaches may be applicable to freshwater systems by using remotely sensed data as a proxy for chlorophyll *a* production (Chegoonian et al., 2023).

Since NDVI is a proxy of primary productivity (Pettorelli et al., 2005), our estimates of DENVar would also relevant to a variety of industries. In agriculture, remotely-sensed mean NDVI has been used to classify crop types (Zhang et al., 2022; Bhatti et al., 2024) and assess crop yield (Roznik et al., 2022). Accounting for spatiotemporal variance in NDVI may help account for the spatiotemporal variation in weather conditions and improve the accuracy, particularly in the case of models of crop yield and even prior to harvest. In the fields of land conervation and management (including forestry), mean NDVI can help assess a habitat's regeneration stage following landscape disturbance (Pickell et al., 2016), including after fire (Lanorte et al., 2014; Zahabnazouri et al., 2025), logging (Chirici et al., 2020), and during restoration efforts (Valtonen et al., 2021). Currently, little work has been done on the change in NDVI variance following recovery efforts or large-scale disturbance, but quantifying such trends may help elucidate the recovery process. Lanorte et al. (2014) found that forests had greater Shannon entropy (Shannon, 1948) in NDVI prior to a fire than after, suggesting that temporal trends in NDVI may be useful metrics for assessing forests' regeneration stage.

Spatiotemporal trends in NDVI stochasticity may also help shed light on the complexity of forest regeration, since younger forests... Such estimates can be particularly useful in boreal ladscapes, where cold weather limits the forest's regeneration (Pickell et al., 2016), and differences in canopy structure can affect the timing and frequency of snowmelt (Law et al., 2001), which will impact water availability and produce markedly different NDVI signatures, with DENVar being higher in areas with more frequent and rapid snow melt. Additionally, frequent disturbance may increase landscape heterogeneity through a wide variety of impacts, including changes in soil microbial communities, water retention, and carbon storage and fluxes (Hartmann et al., 2012; Law et al., 2001). Additionally, the recovery process often differs between logging and fires disturbances (McRae et al., 2001), and both recovery types can be hindered by the interactive effects of climate change (Stoddard et al., 2018).

HERE

4.1 second-last paragraph: Interpreting DENVar: the importance of scale

While we have presented many applications and theoretical backgroung highlighting the importance environmental variance, we have also listed some of the difficulties that come with estimating NDVI variance over time and space.

importange of scale

what counts as a change? predictable vs stochastic changes

smoothness of mean impacts smoothness and size of variance: depends on scale of interest (e.g., animals' ability to respond to, learn, and predict changes in the mean). Changes not attributed to trends in the mean are stochastic

DENVar is a good metric as long as the spatiotemporal scale of interest is similar or larger to that of DENVar. May not work well for small-scale processes such as the movement dynamics of a small rodent, the phenology of a small orchard when spatiotemporal heterogeneity is finer than DENVar's resolution

Calling the metric DENVar is more than just branding: it highlights the importance of using mean and variance estimates at the same spatiotemporal scale.

write a concluding paragraph recalling the uses of DENVar and how it can be quite useful to many

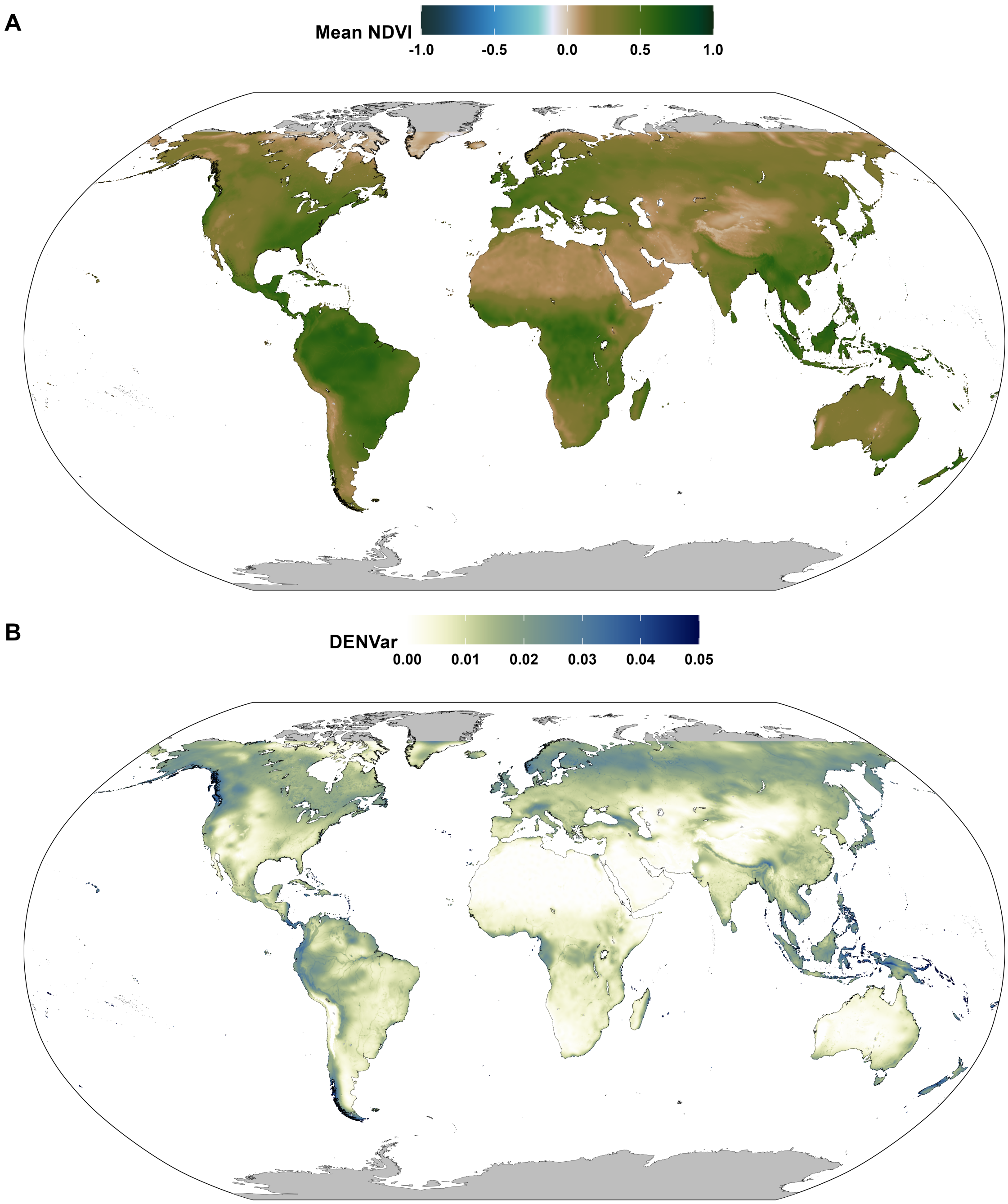

Figure 2: Estimated long-term mean NDVI (A) and DENVar (B).

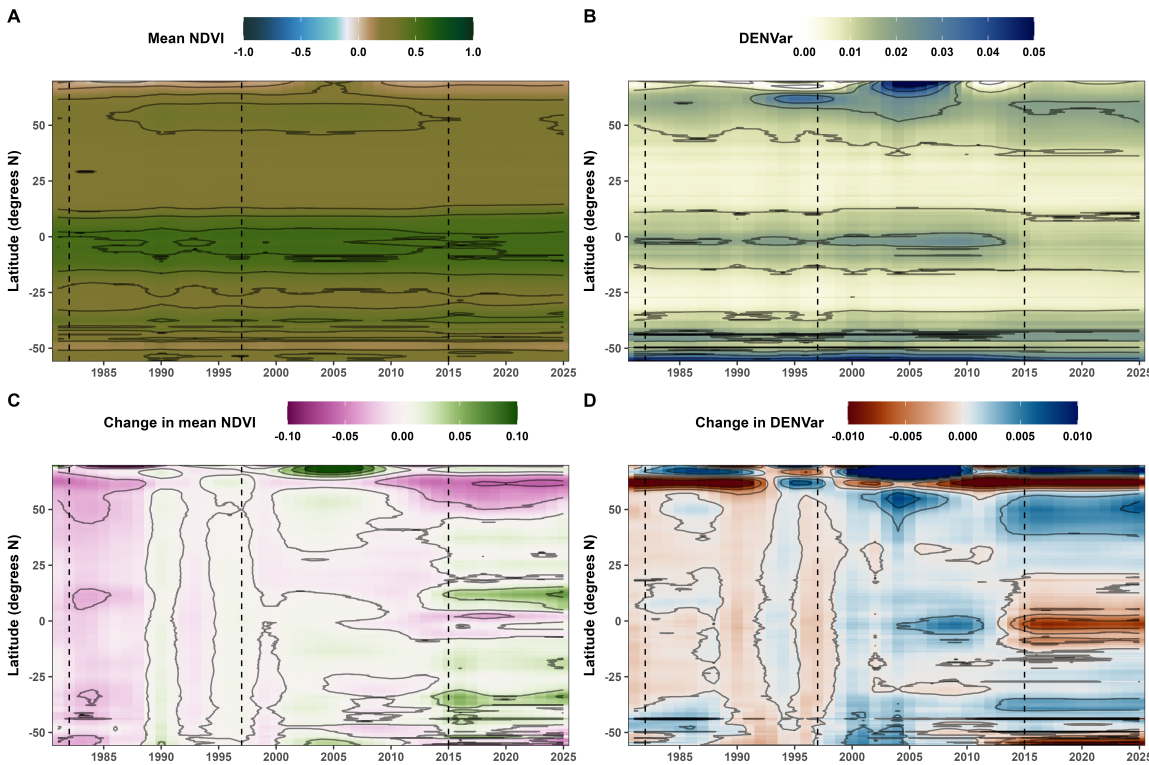

Figure 3: Surface plots of the change in mean NDVI and DENVar over the years. Panels A and B show the estimated averages, while panels C and D show the change in the averages relative to the 1991-2000 average. For ease of readability, DENVar values in panel B have been capped to a maximum of 0.05, while differences in panels C and D have been capped at absolute values of 0.1 and 0.01 ($= 0.1^2$), respectively. Vertical dashed lines indicate years with major El Niño events (Jia et al., 2025).

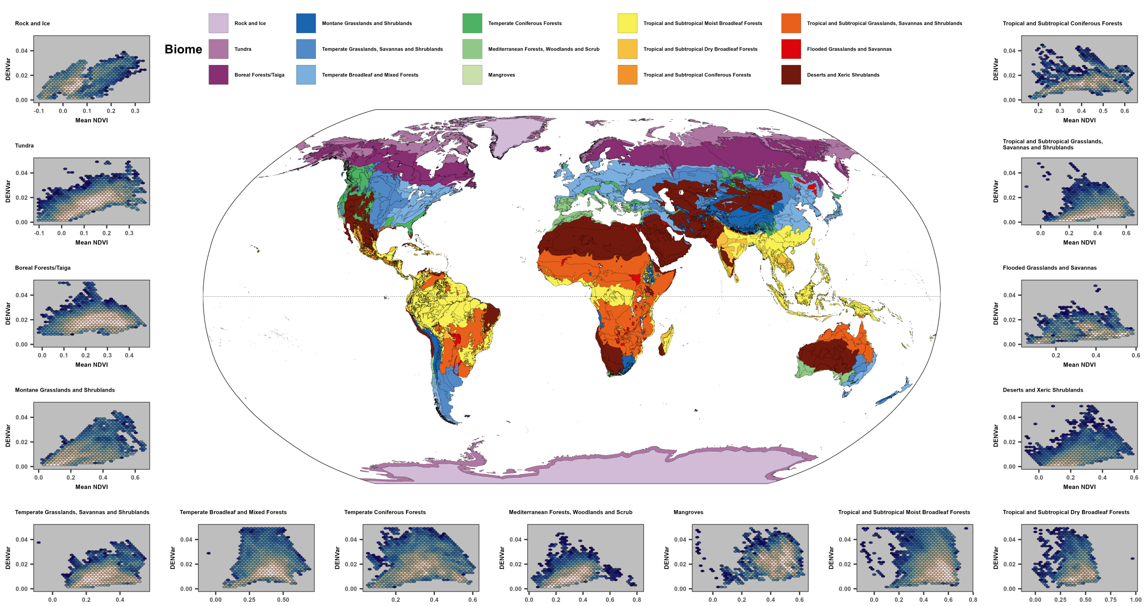

Figure 4: Map of terrestrial biomes along with hex plots of mean NDVI and DENVar for each biome. The fill scale of the hexagonal cells represents the log-10 number of points within the cell, scaled to the biome's total area, with lighter colors indicating higher density.

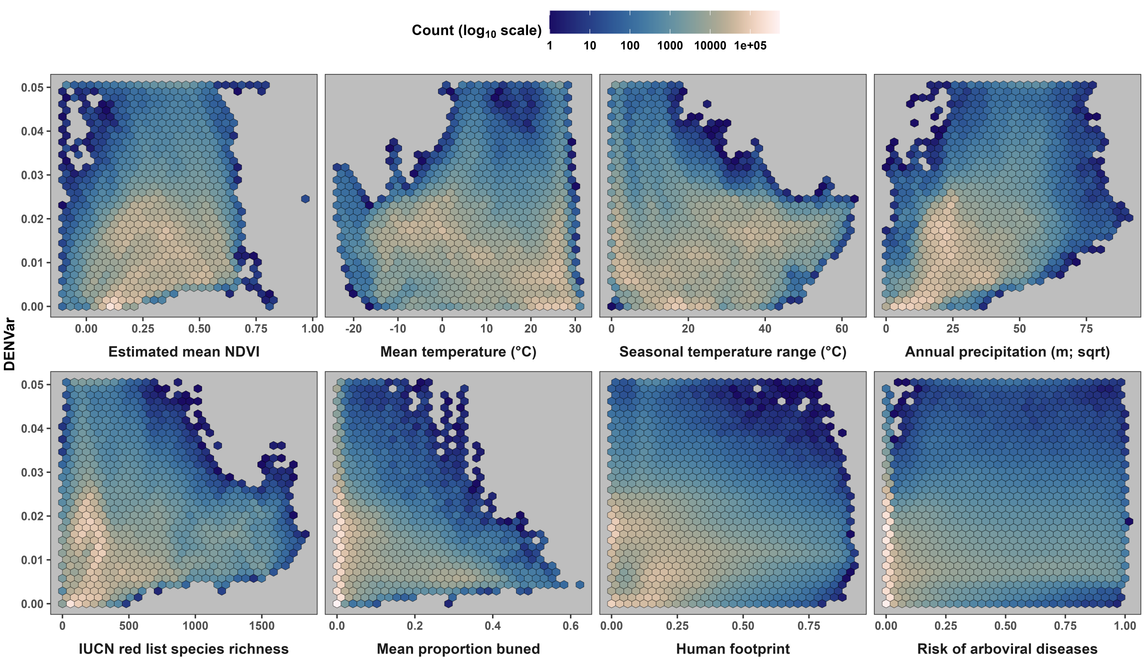

Figure 5: Hex plots of the relationship between DENVar and common ecological metrics. The fill of the hexagonal cells represents the log-10 number of points within the cell. Rowwise, the variables are: long-term mean NDVI (as estimated in this manuscript); mean annual temperature, seasonal temperature range, and annual precipitation (1970-2000) from the work of Fick and Hijmans (2017); IUCN Red List species richness (IUCN, 2025); mean proportion burned (1901-2020; Guo et al., 2025); the human footprint index of Keys et al. (2021); and the risk of dengue, chikungunya, Zika and yellow fever (Lim et al., 2025).

5 Other useful references

- resource stochasticity: (Oestreich et al., 2026)
- READ: Methods for Detecting Early Warnings of Critical Transitions in Time Series Illustrated Using Simulated Ecological Data (https://doi.org/10.1371/journal.pone.0041010)
- READ: Human Impacts Dominate Global Loss of Lake Ecosystem Resilience (https://doi.org/10.1029/2024GL10929
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- increase in variance before eutrophication in lakes: https://doi.org/10.1002/lno.10355
- uses ndvi variance: https://esajournals.onlinelibrary.wiley.com/doi/full/10.1002/ecm.1348
- bro-jorgensen_using_2008: in the dry season preceding the rut topi antelope (*Damaliscus lunatus*) respond to forage availability at the large scale but not the fine scale
- keith_predicting_2008
- chevin_adaptation_2010
- rickbeil_plasticity_2019
- mueller_search_2008
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- tian_evaluating_2015
- huang_commentary_2021
- fan_global_2016
- wang_stochastic_2019
- pease_ecological_2024
- xu_plasticity_2021
- calibrating across sensors: https://ieeexplore.ieee.org/abstract/document/6809983

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Author Contributions

Funding

Conflicts of Interest

The authors declare that there is no conflict of interest regarding the publication of this article.

Code and Data Availability

All code is available on GitHub at https://github.com/QuantitativeEcologyLab/ndvi-stochasticity, including code for downloading the original NDVI rasters directly from the servers they are housed in. Most data are not included due to GitHub's size limitations, but all objects can be created using the code provided.

Supplementary Materials

- Appendix A: Color palettes used and input data.
- Appendix B: Sensitivity analyses.
- Appendix C: Code for fitting the models, along with supporting supplementary figures.

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